

TABLE OF CONTENTS

S-406 METAL DECKING WORKS

	Page No.
1. SCOPE OF WORK.....	S.406-2
2. CODES AND STANDARDS	S.406-2
3. DESIGN	S.406-3
3.1 Deck Types	S.406-3
3.2 Welds, Fasteners And Other Accessories.....	S.406-3
3.3 Construction Loads	S.406-3
4. MATERIAL	S.406-3
4.1 Sheet Material	S.406-3
4.2 Finish.....	S.406-4
5. ABRICATION AND INSTALLATION	S.406-4
5.1 General Provisions	S.406-4
5.2 Deck Position.....	S.406-4
5.3 Opening	S.406-4
5.4 Arrangement	S.406-4
5.5 Alignment	S.406-4
5.6 Closures	S.406-4
5.7 Ventilation	S.406-5
5.8 Field Cutting For Openings	S.406-5
5.9 Fastening.....	S.406-5
5.10 Temporary Supports.....	S.406-6
5.11 Field Touch-Up Painting	S.406-6
6. QUALITY ASSURANCE / QUALITY CONTROL (QA/QC).....	S.406-6

S.406 METAL DECKING WORK

1. SCOPE OF WORK

- 1.1 This specification covers the requirements for metal decking as formwork for cast in place concrete floors and roofs.

Metal decking work shall include Bidder's detailing, fabricating, furnishing, installing and inspecting of the work and shall conform to the requirements of this specification.

- 1.2 Accessories for metal decking shall include but not limited to the following:

- 1.2.1 All fasteners.

- 1.2.2 Closures, including the following:

- End closures
- Side closures
- Column closures

- 1.2.3 Reinforcement under metal decking at columns and framed openings, including miscellaneous structural steel as required for the support of the metal deck at sleeves and unframed openings shall be cut through the deck prior to placing of the concrete. Before commencement of the work, the Bidder shall submit the drawings and calculation sheet to the Owner for approval.

2. CODES AND STANDARDS

- 2.1 The Bidder shall comply with the requirements of Part 4, Subsection 2.5 "Codes and Standards".

- 2.2 The codes standards of the following organizations extracted from the Part 4, subsection 2.5 "Codes and Standards" are specifically applicable to the design, manufacture and testing of the work included in this Specification:

AISI	American Iron and Steel Institute
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
SDI	Steel Deck Institute. U.S.A

JIS Japanese Industrial Standards

3. DESIGN

3.1 Deck Types

3.1.1 Deck type and size shall be designed and detailed by the Bidder and shall be used only for formwork for cast in place concrete.

3.1.2 The deck shall be suitable for the span and the design loads including the construction loads.

3.1.3 The Bidder is not allowed to consider the deck as part of reinforcing steel for the concrete slab.

3.1.4 Minimum thickness of metal deck shall be 1.2 mm and allowable deflection shall conform to the manufacturer's requirement.

3.2 Welds, Fasteners and Other Accessories

Welds, fasteners and other accessories shall be designed and detailed by Bidder in accordance with the applicable requirements of AISI Specifications for the Design of Cold-formed Steel Structural Members and SDI Design Manual for Composite Decks, Form Decks and Roof Decks, and the requirements of the Specification for Welding in Steel Construction, unless otherwise indicated on the Bidder's design drawings.

3.3 Construction Loads

Deck acting as formwork for cast-in-place concrete shall be designed by the Bidder for the wet weight of concrete plus a minimum 150 kg/m² construction live loads. Clause 5.10 indicates the responsibility for temporary support required for any additional construction loads.

The Bidder shall be fully responsible to any failure of the decking work because of miss conduct by the Bidder.

4. MATERIAL

4.1 Sheet Material

ASTM A653, Grade 33 (minimum yield strength 230MPa).

4.2 Finish

Hot-dip galvanized, ASTM A924, Coating Designation G90 (minimum).

5. FABRICATION AND INSTALLATION

5.1 General Provisions

Metal decking, including fasteners and all other accessories, shall be fabricated and installed in accordance with the requirements of this Clause 5, applicable AISI and SDI Specifications and Standards, and in strict accordance with manufacturer's printed instructions unless otherwise indicated.

5.2 Deck Position

For cast-in-place concrete slabs, metal deck shall be installed in inverted position unless otherwise indicated.

5.3 Opening

No opening through deck shall be cut prior to concrete is placed and hardened.

5.4 Arrangement

Where the span of metal deck for cast-in-place concrete slabs is parallel to and directly supported by steel beams, provisions shall be made for full bearing of the rib bottom on steel beams, or the deck shall be cut and side closures provided. If side closures are provided to form a rib bottom, the area of rib profile so formed shall be at least equal to the area of the deck rib profile.

5.5 Alignment

Metal decking for cast-in-place concrete slabs shall be installed so that the ribs line up to permit continuity of bottom reinforcing bars.

5.6 Closures

Metal decking formwork for cast-in-place concrete roofs or floors shall be provided with specific closures, with provisions for reinforcing bars to pass through, and to prevent leakage of concrete during placement at the following location:

5.6.1 Penetrations through metal decking by structural steel or other members.

5.6.2 Ends or discontinuities of metal decking.

5.6.3 Perimeters of openings and cants in metal decking.

5.6.4 All others as indicated on the Bidder's design drawings.

5.7 Ventilation

For metal decking formwork for cast-in-place concrete roofs only, ventilating clips or suitable openings shall be provided for release of moisture between the concrete and the metal decking.

5.8 Field Cutting For Openings

Shall be performed by sawing or nipping; the use of flame cutting, welded electrodes or chiselling will not be permitted.

5.9 Fastening

5.9.1 All metal decks shall be fastened to every supporting structural steel member. Where the ribs of the deck are perpendicular to the supporting steel beams and if the rib spacing of the deck is less than 300 mm, metal deck shall be fastened at the bottom of every other rib. If the rib spacing is 300 mm or more, metal deck shall be fastened at the bottom of every rib.

5.9.2 All ends or butt ends of metal decking shall be fastened to the supporting structural members.

5.9.3 The side laps of metal decking shall be fastened at not more than 500 mm on centres.

5.9.4 When fusion welds are used for fastening, they shall have diameters of not less than 19 mm and shall be free of sharp points or edges.

5.9.5 Where the ribs of metal deck are parallel to and directly supported by steel beams, the bottom of ribs or side closure shall be fastened beams with 19 mm diameter fusion welds at a maximum of 915 mm on centers.

5.9.6 Fasteners and welding shall conform to the requirements of Clause 3.2.

5.10 Temporary Supports

The Bidder shall be responsible for designing, furnishing, installing and removing any temporary supports required to support the additional construction loads resulting from his work, with reference to Clause 3.3

5.11 Field Touch-Up Painting

Field cut edges, scratches etc. in the metal decks shall be field touch-up painted with one coat of zinc rich paint comparable with shop galvanizing.

6. QUALITY ASSURANCE / QUALITY CONTROL (QA/QC)

6.1 Quality control shall conform to the requirements of Part 4, Subsection 2.7

6.2 The quality control procedures shall include but not be limited to the following:

- a. Material certificates
- b. Welding
- c. Non destructive testing
- d. Cleaning

6.3 The quality control documentation shall include but not be limited to the following:

- a. Material certificates
- b. Welding
- c. Field test and inspection reports.

(Last page of S-406)